

Information and Entertainment System



Special Tool(s)

 ST3093-A	Fluke 77-IV Digital Multimeter FLU77-4 or equivalent
 ST2574-A	Flex Probe Kit NUD105-R025D or equivalent

Symptom Charts

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Symptom Chart — General Audio System

Condition	Possible Causes	Action
Crew Chief is inoperative or does not operate correctly	Refer to the Pinpoint Test	GO to Pinpoint Test A.

Pinpoint Test

Pinpoint Test A: Crew Chief Is Inoperative Or Does Not Operate Correctly

Diagnostic Overview

Diagnostics in this manual assume a certain skill level and knowledge of Ford-specific diagnostic practices. Refer to Diagnostic Methods in Section 100-00 for information about these practices.

Refer to Telematics Module and Modem-Antenna.
Refer to Wiring Diagrams Cell 141 for schematic and connector information.

Normal Operation and Fault Conditions

Refer to Crew Chief, Telematics Module, and Modem-Antenna in Information and Entertainment System.

Possible Sources

- Inadequate reception
- Improper installation
- Wiring, terminals, or connectors
- Modem-antenna
- Telematics module

PINPOINT TEST A : CREW CHIEF IS INOPERATIVE OR DOES NOT OPERATE CORRECTLY

A1 CHECK THE RECEPTION

- Check to make sure the location where the vehicle is being diagnosed is within CDMA (Code Division Multiple Access) network coverage. Refer to the [coverage map](#).

Is the location where the vehicle is being diagnosed within CDMA (Code Division Multiple Access) network coverage?

Yes	GO to A2 .
No	Drive the vehicle to a location with adequate CDMA (Code Division Multiple Access) network coverage, and retest the system for normal operation. If the concern is still present, GO to A2 .

A2 CHECK THE FOUR STATUS INDICATORS

- Access the technician service publications website.
- Enter the vehicle VIN .
- Select GO.
- Select the QASIS tab.
- Choose FW Status.
- Start the vehicle and allow it to idle.
- **NOTE:** A system that is operating correctly will have a green status indicator next to four data fields: VIN , Last Contact, Last GPS Fix, and Last CAN Data.

Observe the color of the four status indicators.

Are all four status indicators green?

Yes	The system is operating correctly at this time. The concern may have been caused by the vehicle being outside of CDMA (Code Division Multiple Access) network coverage.
No	Record the date and time stamp for any red status indicator. GO to A3 .

A3 CHECK FOR UPDATED DATE AND/OR TIME STAMPS

NOTE: The telematics module has the capability to accumulate messages if it is unable to transmit due to a loss of CDMA (Code Division Multiple Access) network coverage or a system fault.

- Start the vehicle and allow it to idle for a minimum of 30 seconds.
- Ignition OFF.
- Review the recorded date and time stamp(s) for the red status indicator(s).

Did the date and/or time change for any recorded red status indicator?

Yes	GO to A4 .

No	GO to A5 .
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A4 CHECK FOR ACCUMULATED MESSAGES IN THE TELEMATICS MODULE

- Start the vehicle and allow it to idle.
- Waiting at least 3 minutes between attempts, select the Refresh button on the technician service publications website continuously until the time stamp(s) for the recorded red status indicator(s) no longer update.
- Ignition OFF.
- Observe the color of the four status indicators.

Are all four status indicators green?

Yes	The system is operating correctly at this time. The concern was corrected by allowing the telematics module to send all of the accumulated messages.
No	GO to A5 .

A5 VERIFY PROPER INSTALLATION AND PROVISIONING

- Perform steps A, B, C, and D three times, for a total of three ignition cycles.
 - A. Ignition ON.
 - B. Wait 10 seconds or until the airbag indicator has turned off.
 - C. Start the vehicle and run the engine for a minimum of 30 seconds.
 - D. Ignition OFF.
- Select Refresh on the technician service publications website and observe the color of the four status indicators.

Are all four status indicators green?

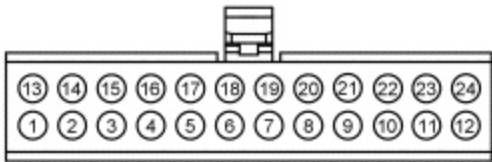
Yes	The system is operating correctly at this time. The concern may have been caused by an incorrectly provisioned or incorrectly installed telematics module or modem-antenna.
No	If the VIN status indicator is red, GO to A8 . If the Last Contact status indicator is red, GO to A6 . If the Last GPS Fix status indicator is red, GO to A9 . If the Last CAN Data status indicator is red, GO to A8 .

A6 CHECK THE TELEMATICS MODULE POWER SUPPLY

- Ignition OFF.
- Disconnect: Telematics Module [C2409](#).
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit

C2409 Pin 12	VB1 (GY/RD)	—	Ground
C2409 Pin 10	IGRUN-C (WH/BN)	—	Ground



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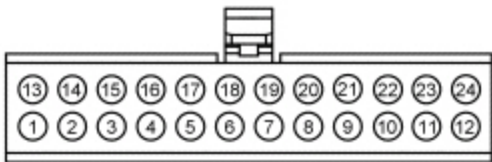
Is the voltage greater than 11 volts?

Yes	GO to A7 .
No	REPAIR the circuit in question.

A7 CHECK THE TELEMATICS MODULE GROUND CIRCUIT

- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C2409 Pin 12	VB1 (GY/RD)	C2409 Pin 1	GND1 (BK/GY)



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Is the voltage greater than 11 volts?

Yes	GO to A8 .
No	REPAIR the circuit.

A8 CHECK THE TELEMATICS MODULE COMMUNICATION CIRCUITS FOR AN OPEN

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C251 Pin 6	VDB04 (WH/BU)	C2409 Pin 15	HSC-1 (WH/BU)
C251 Pin 14	VDB05 (WH)	C2409 Pin 16	HSC-2 (WH)



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Are the resistances less than 3 ohms?

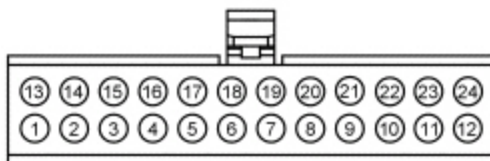
Yes	GO to A9 .
No	REPAIR the circuit in question.

A9 CHECK THE CIRCUITS TO THE MODEM ANTENNA FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: Modem Antenna [C2410](#).
- Ignition ON.
- Measure the **voltage** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C2409 Pin 3	TXD (YE)	—	Ground
C2409 Pin 4	RXD (GN)	—	Ground
C2409 Pin 6	CGND (BK)	—	Ground

C2409 Pin 24	5VTB (RD)	—	Ground
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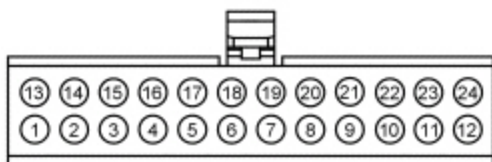
Is any voltage present?

Yes	REPAIR the circuit in question.
No	GO to A10 .

A10 CHECK THE CIRCUITS TO THE MODEM ANTENNA FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C2409 Pin 3	TXD (YE)	—	Ground
C2409 Pin 4	RXD (GN)	—	Ground
C2409 Pin 6	CGND (BK)	—	Ground
C2409 Pin 24	5VTB (RD)	—	Ground



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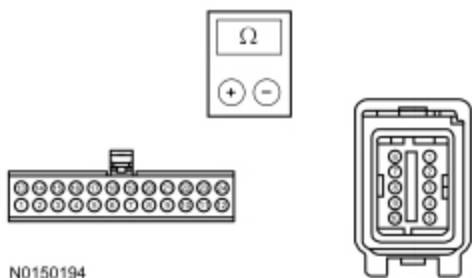
Are the resistances greater than 10,000 ohms?

Yes	GO to A11 .
No	REPAIR the circuit in question.

A11 CHECK THE CIRCUITS TO THE MODEM ANTENNA FOR AN OPEN

• Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C2409 Pin 3	TXD (YE)	C2410 Pin 6	TXD (YE)
C2409 Pin 4	RXD (GN)	C2410 Pin 7	RXD (GN)
C2409 Pin 6	CGND (BK)	C2410 Pin 4	CGND (BK)
C2409 Pin 24	5VTB (RD)	C2410 Pin 8	5VTB (RD)



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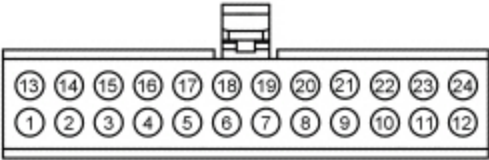
Are the resistances less than 3 ohms?

Yes	GO to A12 .
No	REPAIR the circuit in question.

A12 CHECK THE CIRCUITS TO THE MODEM ANTENNA FOR A SHORT TOGETHER

- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C2409 Pin 3	TXD (YE)	C2409 Pin 4	RXD (GN)
C2409 Pin 3	TXD (YE)	C2409 Pin 6	CGND (BK)
C2409 Pin 3	TXD (YE)	C2409 Pin 24	5VTB (RD)
C2409 Pin 4	RXD (GN)	C2409 Pin 24	5VTB (RD)
C2409 Pin 4	RXD (GN)	C2409 Pin 6	CGND (BK)
C2409 Pin 6	CGND (BK)	C2409 Pin 24	5VTB (RD)



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Are the resistances greater than 10,000 ohms?

Yes	GO to A13 .
No	REPAIR the circuit in question.

A13 ISOLATE THE MODEM-ANTENNA

- Connect: Telematics Module [C2409](#).
- Install a new modem-antenna. Refer to Modem — Antenna.
- Start the vehicle and allow it to idle.
- Select Refresh on the technician service publications website and observe the color of the four status indicators.

Are all four status indicators green?

Yes	The system is operating correctly at this time. The concern was caused by an inoperative modem-antenna.
No	GO to A14 .

A14 CHECK FOR CORRECT TELEMATICS MODULE OPERATION

- Ignition OFF.
- Disconnect and inspect the telematics module connector.
- Repair:
 - Corrosion (clean module pins or install new connectors or terminals)
 - Damaged or bent pins (install new terminals or pins)
 - Pushed-out pins (install new pins as necessary)
- Reconnect the telematics module connector. Make sure it seats and latches correctly.
- Start the vehicle and allow it to idle.
- Select Refresh on the technician service publications website and observe the color of the four status indicators.

Are all four status indicators green?

Yes	The system is operating correctly at this time. The concern may have been caused by module connections. ADDRESS the root cause of any connector or pin issues.
No	CHECK OASIS for any applicable TSBs. If a TSB exists for this concern, DISCONTINUE this test and FOLLOW TSB instructions. If no TSBs address this concern, INSTALL a new telematics module. REFER to Telematics Module.